

# Lecture 16: Taming the Factor Zoo and Shrinking the Cross-Section

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Raul Riva

FGV EPGE

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# Intro

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- Last lecture: shrinkage and double selection
- Today: apply those tools to the factor zoo
- Kozak et al. ([2020](#)) ask how to estimate an SDF in a huge characteristic space
- Feng et al. ([2020](#)) ask whether one new factor still matters after the whole zoo
- Main theme: high dimension changes both estimation and inference

# Why This Lecture?

- Lecture 14: weak test assets and multiple testing made the zoo look dangerous ([Lewellen et al. 2010](#); [Harvey et al. 2016](#))
- Lecture 15: regularization and valid post-selection inference gave us new tools ([Belloni et al. 2014](#))
- Today: two modern answers to the same broad problem
- One estimates a robust high-dimensional SDF
- One tests a candidate factor against a high-dimensional benchmark

- Part I: why Kozak et al. ([2020](#)) reject characteristics-sparse factor models
- Part II: how Feng et al. ([2020](#)) test whether a new factor adds marginal pricing power
- End: compare the two objects, the two methods, and the two contributions

## Part I: Shrinking the Cross-Section

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# The Question

- Can a few characteristics summarize the whole cross-section?
- FF3 and FF5 made that look plausible ([Fama and French 1993, 2015](#))
- Lecture 14 already made us suspicious ([Lewellen et al. 2010](#); [Harvey et al. 2016](#))
- Kozak et al. ([2020](#)) ask: maybe the SDF is not sparse in characteristics at all

# Why Sparse Characteristics Are a Strong Claim

- Present-value identity connected observed prices with expected profitability and investment
- Many characteristics can proxy for profitability, investment, distress, ...
- We don't actually observe expectations
- Sparsity in characteristics means extreme redundancy across anomalies



# SDF Setup

- Start with a  $N \times 1$  vector of excess returns  $R_t$
- $Z_t$ :  $N \times H$  matrix of firm characteristics
- $F_t = Z_{t-1}^\top R_t$  are factor portfolios

They consider an SDF of the following form:

$$M_t = 1 - b_{t-1}^\top (R_t - \mathbb{E}_{t-1}[R_t])$$

where  $b_{t-1} = Z_{t-1}^\top b$

- If a given  $b_j = 0$ , then the  $j$ th characteristic does not enter the SDF at all
- Sparsity in the SDF is equivalent to sparsity of  $b$

Plugging one equation into the other, we can rewrite the SDF as

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- A conditional model for the SDF entails  $\mathbb{E}_{t-1}[M_t R_t] = 0$
- Then it should be the case that

$$\mathbb{E}[M_t F_t] = 0$$

since factors are themselves portfolios of returns!

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- Solving for  $b$  gives:

$$b = \Sigma^{-1} \mu$$

where  $\Sigma = \mathbb{E}[(F_t - \mathbb{E}[F_t])(F_t - \mathbb{E}[F_t])^\top]$  and  $\mu = \mathbb{E}[F_t]$

$$\hat{b}^{naive} = \hat{\Sigma}_1^{-1} \bar{\mu}$$

- This is just the sample analog of the exact formula
- Problem 1: expected returns are noisy
- Problem 2:  $H$  can be large relative to  $T$ , screwing up the covariance estimate
- Result: the estimator overfits means and loads too much on weak directions

## Rotate to Principal Components

$$\Sigma_1 = QDQ^\top, \quad P_t = Q^\top F_t, \quad b_P = D^{-1}\mathbb{E}[P_t]$$

- PCs rank directions by variance
- Low-eigenvalue PCs are statistically fragile
- Directions of low variance generate huge coefficients unless the premia is estimated to be zero
- If we mis-handle them, the estimated SDF becomes wild

## The Economic Prior

- Suppose you know  $\Sigma$
- They assume a Gaussian prior for  $\mu$ :

$$\mu \sim \mathcal{N}\left(0, \kappa^2 \frac{\Sigma^\eta}{\tau}\right), \quad \tau = \text{tr}[\Sigma] = \sum_{j=1}^H d_j$$

where  $d_j$  is the  $j$ -th eigenvalue of  $\Sigma$

- Recall that  $\Sigma^\eta = Q D^\eta Q^\top$
- If  $\eta$  is large, directions of low eigenvalues are taken to zero a priori

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- Recall that  $\Sigma^\eta = Q D^\eta Q^\top$
- If  $\eta$  is large, directions of low eigenvalues are taken to zero a priori
- Notice that  $b \sim \mathcal{N}(0, \kappa^2 \frac{\Sigma^{\eta-2}}{\tau})$
- The variance about  $b$  is huge if  $\eta$  is too small
- They pick  $\eta = 2$



# The Shrinkage Estimator

- We get a spherical prior on coefficients:  $b \sim \mathcal{N}\left(0, \frac{\kappa^2}{\tau} I_H\right)$
- Now they assume factors come from an i.i.d Gaussian distribution
- This makes you able to derive the posterior distribution of  $b$  given the *observed* mean of factors:

$$\hat{b} = (\Sigma + \gamma I_H)^{-1} \bar{\mu}$$

where  $\gamma = \frac{\kappa^2}{T \cdot \tau}$

- This looks like something you saw last class, right?

## Penalized Interpretation

$$\hat{b} = \arg \min_b \{ (\bar{\mu} - \Sigma b)^\top \Sigma^{-1} (\bar{\mu} - \Sigma b) + \gamma b^\top b \}$$

- That estimator is equivalent to the argmin of the above objective function
- Recall that  $\|b\|_2^2 = b^\top b$
- This is like Ridge but with a non-standard loss function

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- Another equivalent optimization problem is

$$\hat{b} = \arg \min_b \{(\bar{\mu} - \Sigma b)^\top (\bar{\mu} - \Sigma b) + \gamma b^\top \Sigma b\}$$

- This is like Ridge but with a different penalty
- $b^\top \Sigma b$  is the maximal Sharpe ratio attained. Why?

## Why Add an $\ell_1$ Penalty?

$$\hat{b} = \arg \min_b \left\{ (\bar{\mu} - \Sigma_1 b)^\top \Sigma^{-1} (\bar{\mu} - \Sigma_1 b) + \gamma_2 b^\top b + \gamma_1 \sum_{j=1}^H |b_j| \right\}$$

- $\ell_1$  can drop useless factors
- But pure lasso is bad with correlated predictors
- And characteristic portfolios are very correlated
- This is like the Elastic Net, with a non-standard loss function
- Their main empirical message:  $\ell_2$  matters much more than sparse selection

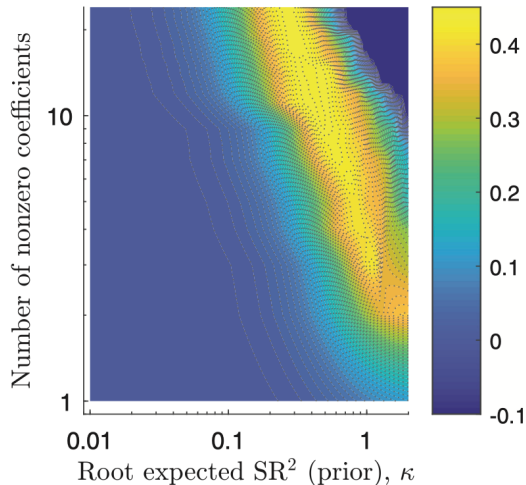
## How To Pick Penalty Parameters?

- They use  $K$ -fold cross-validation
- They use  $K = 3$  and split the data in a contiguous fashion
- After estimating the SDF on  $K - 1$  folds, they compute the out-of-sample  $R^2$  on the holdout fold

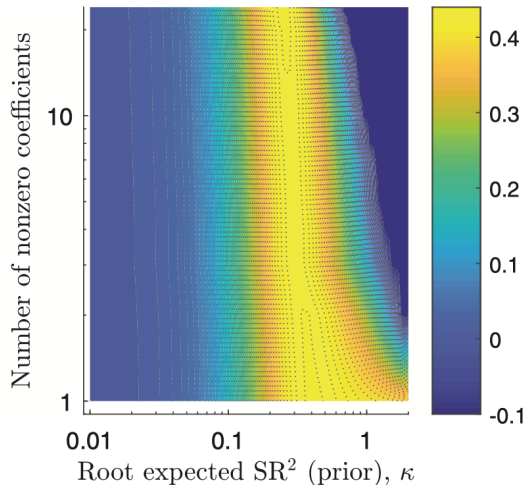
$$R_{OOS}^2 = 1 - \frac{(\bar{\mu}_K - \bar{\Sigma}_K \hat{b})^\top (\bar{\mu}_K - \bar{\Sigma}_K \hat{b})}{\bar{\mu}_K^\top \bar{\mu}_K}$$

- Keep rotating and average values

## FF25 as a Sanity Check

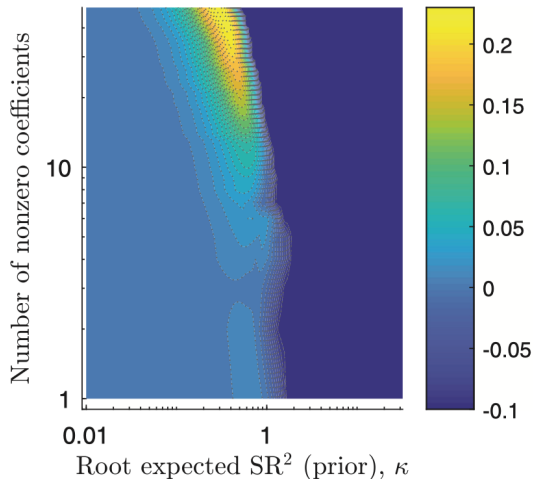


(a) Raw Fama-French 25 portfolios

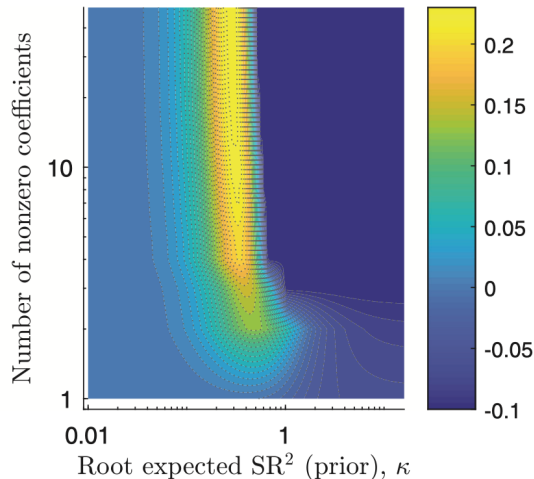


(b) PCs of Fama-French 25 portfolios

## 50 Anomalies From The Literature

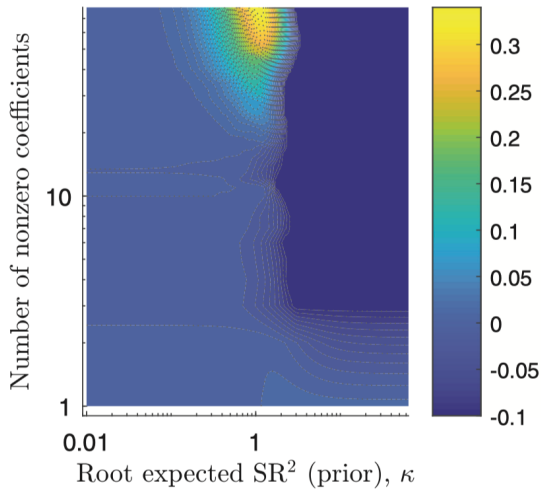


(a) Raw 50 anomaly portfolios

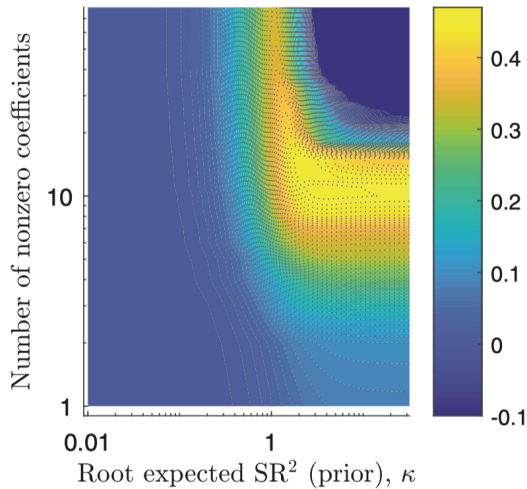


(b) PCs of 50 anomaly portfolios

# WRDS Fincial Ratios: 70 Portfolios



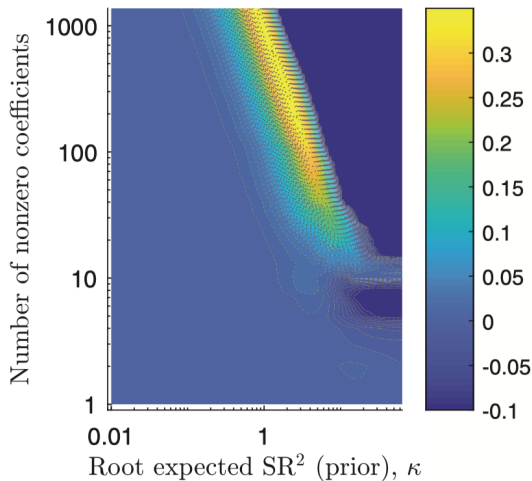
(a) Raw WFR portfolios



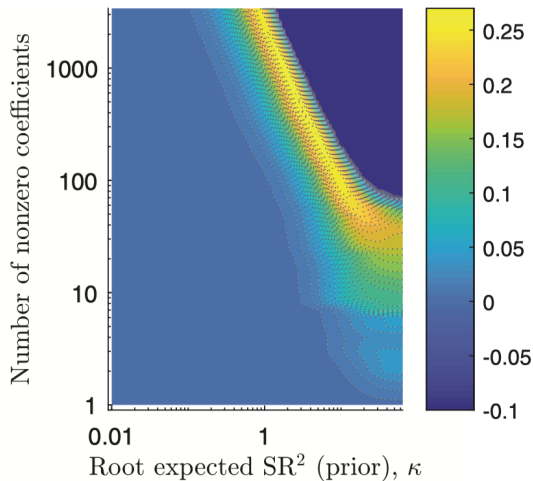
(b) PCs of WFR portfolios



# Considering Non-Linearities: Characteristics and Their Powers



(a) 50 anomalies



(b) WFR portfolios

- Not another factor paper: it is a paper about how to estimate the SDF in high dimension
- It rejects the hope that a tiny set of characteristics is enough
- It gives a formal role to shrinkage in empirical asset pricing
- One line summary: the SDF is not sparse in characteristics, but it is still shrinkable
- Their view: Asset Pricing is about many *weak* factors...

Questions?

## Part II: Taming the Factor Zoo

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# The Question

- Suppose someone proposes a new factor today
- How should we test it against the whole existing zoo?
- FF3 or FF5 are convenient benchmarks, but also arbitrary
- Feng et al. ([2020](#)) propose a conservative test on purpose

$$m_t = \gamma_0^{-1} (1 - \lambda_g^\top g_t - \lambda_h^\top h_t)$$

$$\mathbb{E}[r_t] = \iota_n \gamma_0 + C_g \lambda_g + C_h \lambda_h$$

- $g_t$ : factor we want to test
- $h_t$ : potentially huge set of control factors
- The goal is inference on  $\lambda_g$

$$\mathbb{E}[r_t] = \iota_n \gamma_0 + \beta_g \gamma_g + \beta_h \gamma_h$$

- Risk premia and SDF loadings are not the same thing
- A factor can earn a premium just because it is correlated with true pricing factors
- Feng care about whether the factor enters the SDF
- So the target is  $\lambda_g$ , not just average return

# Why Standard Two-Pass Breaks

- If  $p$  is small, ordinary two-pass methods are fine
- But in the zoo,  $p$  can be comparable to  $n$  or  $T$
- Using all factors is infeasible or very noisy
- Using FF3 or FF5 alone creates omitted-variable bias



## Sparsity as a Working Assumption

- They assume the relevant control set is approximately low-dimensional
- This is the same broad “bet on sparsity” logic as Lecture 15
- But the benchmark is not fixed ex ante
- The benchmark must be selected from a large set of candidates

# Why Plain LASSO Is Not Enough

- LASSO is good for prediction
- Inference fails if an omitted control is correlated with the factor of interest
- So selection mistakes can create omitted-variable bias
- This is exactly the Belloni-Chernozhukov-Hansen logic from last lecture ([Belloni et al. 2014](#))

## Double Selection: Step 1

$$\min_{\gamma, \lambda} \left\{ n^{-1} \left\| \bar{r} - \iota_n \gamma - \hat{C}_h \lambda \right\|_2^2 + \tau_0 n^{-1} \|\lambda\|_1 \right\}$$

- First LASSO selects controls that help explain the cross-section
- This is the “pricing ability” margin
- It looks like the natural factor-selection step
- But it is not enough for valid inference on  $\lambda_g$

$$\min_{\xi_j, \chi_j} \left\{ n^{-1} \left\| \hat{C}_{g,j} - \iota_n \xi_j - \hat{C}_h \chi_j^\top \right\|_2^2 + \tau_j n^{-1} \|\chi_j\|_1 \right\}$$

- Second LASSO searches for controls whose exposures line up with exposures to  $g_t$
- This is the omitted-variable-bias margin
- It keeps factors that may look weak for pricing but are dangerous to omit

$$(\hat{\gamma}_0, \hat{\lambda}_g, \hat{\lambda}_h) = \arg \min_{\gamma_0, \lambda_g, \lambda_h} \left\{ \left\| \bar{r} - \iota_n \gamma_0 - \hat{C}_g \lambda_g - \hat{C}_h \lambda_h \right\|_2^2 : \lambda_{h,j} = 0 \text{ for } j \notin \hat{I} \right\}$$

- Refit by OLS after taking the union of selected controls
- This removes the regularization bias from LASSO
- Same structure as post-double-selection in Lecture 15

## Why This Delivers Valid Inference

- First stage protects pricing relevance
- Second stage protects against omitted-variable bias
- Post-selection OLS restores an interpretable slope estimate
- So inference on  $\lambda_g$  remains valid even with imperfect model selection

- 150 monthly factors from 1976 to 2017
- 15 standard public factors plus 135 long-short proxies
- 750 characteristic-sorted test portfolios
- The large test set directly answers the LNS critique ([Lewellen et al. 2010](#))

Placeholder for Table I  
Testing factors introduced in 2012-2016

- The punchline: most new factors look redundant or useless relative to the old zoo
- A few survive, especially profitability-style factors



Placeholder for Figure 1  
First-stage factor selection rates across subsamples

- Selected models can vary a lot
- That is exactly why “oracle” intuition is dangerous
- Model uncertainty is not a nuisance detail here

Placeholder for Figure 2  
Robustness of  $t$ -statistics to tuning parameters

- The selected benchmark changes with the penalty
- Yet some core conclusions are fairly stable
- Important lesson: benchmark uncertainty and inference must be handled together

## Screening the Zoo Over Time

- They also test factors recursively as they arrive
- In that exercise, many proposed factors would have been screened out at birth
- This fits the broader worry that published anomalies often decay ([McLean and Pontiff 2016](#))
- The procedure is meant to discipline discovery, not encourage more fishing

- It turns Lecture 15's double-selection logic into an asset-pricing test
- It replaces arbitrary benchmarks with selected but inference-valid benchmarks
- It responds directly to LNS and complements HLZ ([Lewellen et al. 2010](#); [Harvey et al. 2016](#))
- It makes factor discovery conservative on purpose

Questions?

## Part III: Putting Them Together

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## Same Problem, Different Objects

- Kozak: estimate a robust high-dimensional SDF
- Feng: test one candidate factor conditional on many controls
- Kozak ask which span matters for pricing
- Feng ask whether one new direction still adds anything beyond the existing span

- Lecture 12-13 built factor models
- Lecture 14 showed the zoo and the weakness of old testing standards
- Lecture 15 gave us shrinkage and valid post-selection inference
- Lecture 16: modern empirical asset pricing means regularization plus tough benchmarks
- A factor is not interesting because it has a story; it must survive high-dimensional comparison



## Readings and References i

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